

Carleson anatomy at the spectral edge

PRIME.RDAGGER.CARLESON_EDGE BAND.01 — round 482

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Abstract

The r460 spectral measure is measured on every dyadic scale $2^{-1}, \dots, 2^{-20}$ for the interval-certified selected windows $k = 5, \dots, 16$. A finite uniform envelope exists:

$$F_k(t) \leq 0.666528 t^{1.25} \quad (0 < t \leq \tfrac{1}{2}, \quad 5 \leq k \leq 16),$$

where the constant is taken over every spectral jump, not just the dyadic samples. This is not a cofinal theorem. More decisively, even the separately optimized Carleson envelope of each exact finite spectrum gives a sufficient-rule value between 1.055000 and 2.700019: it closes none of the twelve windows whose direct $q_k^\dagger$ is already below one. Christoffel and local-Plancherel majorants lose the measured edge alignment; the only scramble-sensitive large-sieve formulation is the target projector inequality itself, with no classical transfer through the data-dependent orthogonal-polynomial basis. The verdict is

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STUCK(SR_FAILS_FINITE_CENSUS;
RESTRICTED_LARGE_SIEVE_IS_TARGET;
STRICT_F_SCRAMBLE_PREMISE_FALSE).
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No RH claim and no anti-RH claim.

1 Object and the sufficient rule

For the production selected window

$$a_k = 2^k, \quad r_k = \lfloor \sqrt{k} \rfloor, \quad m_k = k2^{r_k} - 1, \quad \Delta_k = 2^{-r_k} \log 2,$$

let C_k be the r460 negative-sampling contraction and $u_k = b_k/\sqrt{B_k}$ the budget-normalized border vector in the μ_k -orthonormal polynomial basis. If $C_k e_{k,j} = \lambda_{k,j} e_{k,j}$, put

$$d_{k,j} = 1 - \lambda_{k,j}, \quad w_{k,j} = |\langle u_k, e_{k,j} \rangle|^2, \quad F_k(t) = \sum_{0 < d_{k,j} \leq t} w_{k,j}.$$

Thus $q_k^\dagger = \sum_j w_{k,j}/d_{k,j} = \int (1 - \lambda)^{-1} d\sigma_k(\lambda)$. All MAIN eigenvalues in this census lie below one.

Lemma 1 (Carleson sufficient rule). *If $F_k(t) \leq Ct^{1+\varepsilon}$ for $0 < t \leq \eta$, then*

$$q_k^\dagger \leq \frac{\sigma_k([0, 1 - \eta])}{\eta} + C(1 + \varepsilon^{-1})\eta^\varepsilon. \tag{SR}$$

In particular, a strict upper bound below one implies $q_k^\dagger < 1$.

Proof. With $x = 1 - \lambda$ and $F(t) = \sigma(\{0 < x \leq t\})$, Stieltjes integration by parts gives

$$\int_{(0, \eta]} \frac{dF(x)}{x} = \frac{F(\eta)}{\eta} + \int_0^\eta \frac{F(x)}{x^2} dx \leq C\eta^\varepsilon + \frac{C}{\varepsilon}\eta^\varepsilon.$$

On $x \geq \eta$, $x^{-1} \leq \eta^{-1}$. At an atom on the splitting boundary the displayed closed intervals may double-count; the inequality remains valid. The computation below minimizes between adjacent atoms, so it does not rely on equality at a boundary. $\square$

2 Dyadic anatomy

Tables 1 and 2 give the full requested curve. Each entry is $F_k(2^{-j})$. Zeros are actual empty edge bands, not rounded positive values.

Table 1: Dyadic edge mass, scales $j = 1, \dots, 10$.

$k \backslash j$	1	2	3	4	5	6	7	8	9	10
5	7.95e-3	7.95e-3	0	0	0	0	0	0	0	0
6	1.05e-2	1.75e-6	1.75e-6	1.75e-6	0	0	0	0	0	0
7	2.12e-3	1.18e-3	5.55e-4	5.55e-4	5.55e-4	5.55e-4	0	0	0	0
8	8.46e-2	1.82e-3	1.56e-3	2.76e-4	2.76e-4	2.76e-4	2.76e-4	0	0	0
9	9.03e-2	5.38e-2	1.45e-3	6.94e-5	6.94e-5	3.23e-5	1.22e-5	1.22e-5	1.22e-5	1.22e-5
10	7.78e-2	9.45e-3	7.88e-4	3.72e-4	8.35e-5	4.90e-5	4.90e-5	4.90e-5	2.74e-6	2.74e-6
11	1.66e-1	1.79e-3	1.67e-3	5.22e-5	3.81e-5	3.81e-5	9.85e-6	9.85e-6	9.85e-6	2.44e-6
12	1.55e-1	4.65e-2	1.45e-3	1.22e-3	3.43e-5	2.51e-5	1.19e-5	9.16e-6	9.16e-6	9.16e-6
13	1.01e-1	4.69e-2	8.15e-4	4.75e-4	1.07e-4	1.68e-5	1.31e-5	5.14e-6	5.14e-6	3.26e-6
14	1.20e-1	8.05e-2	5.65e-4	3.87e-4	3.50e-4	1.08e-5	1.08e-5	4.25e-6	3.28e-6	3.28e-6
15	1.03e-1	6.48e-2	1.84e-2	2.71e-4	1.67e-4	1.16e-4	1.16e-5	1.14e-5	3.43e-6	1.95e-6
16	1.36e-1	1.11e-2	6.41e-4	4.00e-4	6.14e-5	2.22e-5	8.37e-6	4.39e-6	7.03e-7	4.82e-7

Table 2: Dyadic edge mass, scales $j = 11, \dots, 20$.

$k \backslash j$	11	12	13	14	15	16	17	18	19	20
5	0	0	0	0	0	0	0	0	0	0
6	0	0	0	0	0	0	0	0	0	0
7	0	0	0	0	0	0	0	0	0	0
8	0	0	0	0	0	0	0	0	0	0
9	0	0	0	0	0	0	0	0	0	0
10	2.74e-6	0	0	0	0	0	0	0	0	0
11	2.44e-6	2.44e-6	0	0	0	0	0	0	0	0
12	6.73e-7	6.73e-7	6.73e-7	0	0	0	0	0	0	0
13	3.26e-6	4.73e-7	4.73e-7	4.73e-7	0	0	0	0	0	0
14	1.29e-6	1.29e-6	2.45e-7	2.45e-7	2.45e-7	0	0	0	0	0
15	1.95e-6	6.91e-7	6.91e-7	1.21e-7	1.21e-7	1.21e-7	0	0	0	0
16	4.07e-7	3.23e-7	5.18e-8	4.96e-8	4.96e-8	7.27e-9	7.27e-9	0	0	0

2.1 Fits, AIC, and drops

The regression uses only positive dyadic values: $\log F = \log C + (1 + \varepsilon) \log t$. “drop” is the disclosed number of zero dyads. ΔAIC compares the free exponent (two parameters) to exponent one (one parameter); negative favours the free exponent. This regression is descriptive. The last column is instead the rigorous envelope constant for the frozen spectrum,

$$C_k^{(.25)} = \max_{\substack{d_{k,j} \leq 1/2 \\ m \leq j}} \frac{\sum_{i \leq m} w_{k,i}}{d_{k,m}^{1.25}},$$

taken at every exact jump after ordering the positive gaps.

Pooling all 127 positive observations gives $C_{\text{reg}} = 0.0190020$, $\varepsilon_{\text{reg}} = 0.274663$, $\text{AIC} = 137.991$ and $\Delta\text{AIC} = -23.696$. This pooled slope does not remove the cross-window instability: the individual exponent excess ranges from -0.631 to $+2.765$. On the exact finite spectra, not merely the dyadic grid, the uniform pair

$$(C, \varepsilon, \eta) = (0.666527842, 0.25, 0.5)$$

works for all $k = 5, \dots, 16$. Finiteness makes existence of some envelope unsurprising; this row says nothing about infinitely many k .

3 The SR balance

For each k and each $\varepsilon \in (0, 4]$ the code orders all exact spectral gaps. Between two consecutive gaps the edge constant is fixed, so $A/\eta + B\eta^\varepsilon$ is minimized analytically at its stationary point or an interval

Table 3: Power fits and exact finite-spectrum envelopes.

k	n	drop	C_{reg}	ε_{reg}	ΔAIC	$C_k^{(.25)}$
5	2	18	—	—	—	0.071253
6	4	16	0.010469	2.765281	-0.371	0.043504
7	6	14	0.001925	-0.630964	-10.397	0.123841
8	7	13	0.026820	0.169318	1.689	0.222095
9	10	10	0.048175	0.476566	-1.316	0.307378
10	11	9	0.035956	0.382989	-4.251	0.189380
11	12	8	0.012824	0.217173	0.509	0.522558
12	13	7	0.052105	0.417614	-4.372	0.523342
13	14	6	0.027714	0.299983	-2.768	0.327817
14	15	5	0.027108	0.287465	-2.409	0.666528
15	16	4	0.051598	0.329506	-6.315	0.620111
16	17	3	0.018161	0.360118	-10.640	0.327005

endpoint. A deterministic grid locates the ε basin and a bounded scalar minimization refines it.

Table 4: Best separate finite-spectrum SR balance. Margin is 1—SR.

k	η	ε	C	far	edge	SR	margin
5	1.000000	0.877030	0.214041	0.620952	0.458092	1.079044	-0.079044
6	0.882140	2.473420	0.548160	0.490507	0.564493	1.055000	-0.055000
7	0.617552	0.214229	0.106085	1.006840	0.542290	1.549130	-0.549130
8	0.626667	0.311907	0.235612	0.735956	0.856583	1.592540	-0.592540
9	0.999123	0.483865	0.425816	0.231699	1.305291	1.536990	-0.536990
10	0.677853	0.448722	0.218131	0.843784	0.591497	1.435280	-0.435280
11	0.286163	0.224233	0.098080	1.597035	0.404484	2.001520	-1.001520
12	0.375508	0.410194	0.356707	0.970016	0.820565	1.790581	-0.790581
13	0.789145	0.361612	0.381615	0.384508	1.319005	1.703513	-0.703513
14	0.154276	0.103930	0.029049	2.445934	0.254085	2.700019	-1.700019
15	0.512828	0.410749	0.838622	0.373879	2.189326	2.563205	-1.563205
16	0.972894	0.441995	0.416970	0.158621	1.343929	1.502550	-0.502550

Every direct value is live, $q_k^\dagger \in [0.852160, 0.945017]$, but every optimized SR majorant is above one. Thus the loss is in replacing a many-step spectral distribution by one power envelope, not in the underlying finite positivity. The r460 two-band calculation and the present calculation share the same failure: a coarse far-field denominator and a separate edge majorant discard the detailed placement of the atoms.

4 World controls: strict edge versus dangerous edge

The reviewer premise needs one correction. The r467 factor $28.5\text{--}5.3 \cdot 10^5$ concerned the normalized overlap with the dangerous top eigenvector. It was not a measurement of $F_k^{\text{SCR}}(t) = \sigma_k^{\text{SCR}}([1-t, 1])$. SCRAMBLE and ARCH generally have eigenvalues at or above one, which strict F excludes. Define the control diagnostic

$$D_k(t) = \sigma_k([1-t, \infty)).$$

It agrees with $F_k(t)$ in the MAIN world but also records a supercritical instability. At $t = 1/2$, the normalized fractions $D/\|u\|^2$ are as follows.

The dangerous fraction is larger under SCRAMBLE for every k , and ARCH is at least 0.9993. The world test therefore confirms the r467 joint-alignment diagnosis. But strict $F^{\text{SCR}}(1/2) < F^{\text{M}}(1/2)$ at $k = 6, 7$: its mass moved through $\lambda = 1$. A proof mechanism must be sensitive to this distinction; treating zero strict mass as a good scrambled control is a category error.

Table 5: Strict masses and normalized dangerous masses at $t = 1/2$.

k	F^M	F^{SCR}	F^{ARCH}	D^M/γ_M	$D^{\text{SCR}}/\gamma_{\text{SCR}}$	$D^{\text{ARCH}}/\gamma_{\text{ARCH}}$
5	7.95e-3	3.12e0	0	0.0095	0.2985	0.9993
6	1.05e-2	0	0	0.0133	0.2297	1.0000
7	2.12e-3	0	0	0.0034	0.0503	1.0000
8	8.46e-2	5.74e3	8.06e-3	0.1551	0.4992	1.0000
9	9.03e-2	2.52e4	0	0.1436	0.8342	1.0000
10	7.78e-2	1.25e0	2.32e-5	0.1181	0.9579	1.0000
11	1.66e-1	2.67e-1	0	0.3477	0.9957	1.0000
12	1.55e-1	4.18e0	0	0.3558	0.9766	1.0000
13	1.01e-1	1.61e-1	0	0.1925	0.9978	1.0000
14	1.20e-1	1.16e0	0	0.3182	0.9999	1.0000
15	1.03e-1	3.05e0	9.19e-5	0.2953	0.9839	1.0000
16	1.36e-1	7.85e-1	7.85e-6	0.2444	1.0000	1.0000

5 The three proposed mechanisms

5.1 Christoffel functions

Let $\mathcal{P}_{<N_k}$ carry the $L^2(\mu_k)$ norm and let ℓ_k be the normalized border functional represented by u_k . The Christoffel/RKHS inequality is

$$|\ell_k(p)|^2 \leq \|\ell_k\|_{\mathcal{P}_{<N_k}^*}^2 \|p\|_{\mu_k}^2 = \|u_k\|^2 \|p\|_{\mu_k}^2. \quad (\text{Chr})$$

Pointwise Christoffel bounds followed by a triangle inequality over the border atoms are weaker. Applied to the top unit eigenvector, (Chr) exceeds the measured squared overlap by factors 105 through $7.64 \cdot 10^7$. More importantly it contains no t . Classical endpoint asymptotics for regular fixed measures do not apply uniformly to this moving, discrete, source-dependent μ_k ; proving the needed regularity is the retained orthogonal-polynomial/RH problem.

Scramble and alias. The numerical Christoffel kernel changes under scramble only because the entire measure changes; the abstract bound does not retain the arithmetic node placement and gives no favourable direction. This is r469 candidate (g), the Loewner/RKHS majorant, and it again separates the extremal data from its edge mode. Verdict: **KILL**.

5.2 Discrete large sieve

Put $E_k(t) = \mathbf{1}_{[1-t,1)}(C_k)\mathbb{R}^{N_k}$. The exact restricted inequality required is

$$\sum_{e_{k,j} \in E_k(t)} |\ell_k(e_{k,j})|^2 = \|E_k(t)u_k\|^2 \leq Ct^{1+\varepsilon}. \quad (\text{RLS})$$

The literal arithmetic positions enter twice:

$$\{\log n\} \longrightarrow c_k^P \longrightarrow (\mu_k, \nu_k) \longrightarrow \{p_{k,i}\} \longrightarrow (C_k, u_k).$$

This joint dependence is exactly why (RLS) passes the methodological scramble filter. Numerically, at $\varepsilon = 0.25$, the sharp finite constants on $t \leq 1/2$ are 0.0435–0.6665 (apart from $k = 5$, 0.0713); the factor to the binding measurement is one by construction. That is a calibration, not a proof.

A classical discrete large sieve controls an exponential sum in terms of node separation and coefficient energy *before* this nonlinear Stieltjes/orthogonalization map. No known theorem transfers such a bound through the MAIN-dependent basis while preserving a positive power of the singular-value defect t . Taking the supremum over unit $p \in E_k(t)$ shows that (RLS) is exactly the Carleson target, not a strict reduction of it. Replacing it by a raw-node large sieve repeats the r467 source-to-border-basis gap. Verdict: **BEST TYPING, BUT TARGET-EQUIVALENT; OPEN**.

5.3 Local Plancherel

Let Q_k project onto the lowest quarter of the orthonormal DCT modes in coefficient degree and let $P_k(t)$ be the edge projector. The exact elementary split

$$\|P_k(t)u_k\| \leq \|Q_k u_k\| + \|P_k(t)(I - Q_k)\| \|u_k\| \quad (\text{P1})$$

is a local-Plancherel budget. At $t = 2^{-8}$ and $k = 9, \dots, 16$ its square exceeds the measured $F_k(t)$ by factors $5.52 \cdot 10^3$ – $3.59 \cdot 10^4$. The low-frequency character of individual top modes does not imply a small operator leakage for the whole edge subspace, and the triangle split loses their phase relation with u_k .

Scramble and alias. P_k and u_k both change under scramble, but inequality (P1) treats their two contributions separately. It is the finite-coordinate form of r469 candidate (c), whose missing edge symbol/Riemann–Hilbert asymptotics were already identified, and of the r467 Abel separation. Verdict: **KILL AT THE 10³ GATE.**

6 Mechanism ranking and closure status

Table 6: Reviewer-candidate ranking.

candidate	sharpness	scramble judgment		status / alias
Christoffel	$105\text{--}7.64 \cdot 10^7$	abstract blind	bound	KILL; r469(g), separate-supremum majorant
restricted large sieve	factor 1 tautologically	yes, literal nodes in C_k, u_k		best typing; (RLS) equals target, raw-node version has r467 gap
local Plancherel	$5.52 \cdot 10^3\text{--}3.59 \cdot 10^4$	objects change, estimate splits them		KILL; r469(c), r467 Abel class

Closing status, verbatim. STUCK(SR_FAILS_FINITE_CENSUS; RESTRICTED_LARGE_SIEVE_IS_TARGET; STRICT_F_SCRAMBLE_PREMISE_FALSE). The finite uniform Carleson envelope is real, but an independently optimized SR envelope is > 1 on every certified window. The data do not refute a uniform Carleson law on infinitely many selected k ; they do show that the measured constants and a single power envelope do not recover even the known finite inequalities. The only candidate that retains the arithmetic/eigenbasis correlation is (RLS), which is the desired projector estimate itself and has no applicable classical literature theorem. Hence neither SR_PROVED nor CARLESON_REDUCED nor CARLESON_REFUTED is justified.

7 Numerical sealing and kill table

The probe reconstructs the r463-faithful active source $n < a_k$ and reproduces the r459/r468 $q^\dagger$ pins. The curve record has SHA-256 prefix `e45622d390abf932`. For the frozen binary64 symmetric matrix, if $E\hat{\Lambda}E^T$ is the computed decomposition, Weyl’s theorem encloses every ordered eigenvalue within $\|CE - E\hat{\Lambda}\|_2 \leq 2 \cdot 10^{-15}$. The closest dyadic threshold is more than $2.5 \cdot 10^8$ such radii away. Independent dps-70 diagonalization at $k = 5, 9, 12, 16$ changes the top eight eigenvalues by at most $4.45 \cdot 10^{-16}$.

These are validated enclosures for the frozen matrix and certify every dyadic membership decision. They are not a fresh interval propagation of all source logs through every eigenvector overlap; the rigorous r465/r468 interval certificates remain the authority for $q^\dagger < 1$. Accordingly the F ordinates are sealed high-precision census values, not promoted interval theorems.

Table 7: Kill and honesty gates.

gate	status	reason
production object / $k = 5..16$	PASS	r463 active source; q pins reproduced
dyadic curves / determinism	PASS	240 values; sealed curve hash
AIC / drops	PASS	zeros disclosed; free vs exponent-one comparison
finite uniform pair	PASS	exact jumps, not only dyads
$SR < 1$	KILL	best 1.055000; no finite closure
SCRAMBLE strict F	KILL as premise	two reversals from $\lambda \geq 1$
SCRAMBLE dangerous mass	PASS	larger on all twelve worlds
Christoffel	KILL	no t ; up to $7.64 \cdot 10^7$ loose
local Plancherel	KILL	minimum factor $5.52 \cdot 10^3$
large sieve	OPEN / TARGET	(RLS) is exactly the projector embedding
33-route alias check	PASS	31 dossier routes plus r467/r468 checked
edge dps / membership	PASS	Weyl radius and dps-70
source-to- F interval weights	NOT PROMOTED	finite census only
cofinal quantifier	OPEN	no extrapolation from $k \leq 16$
no RH claim	PASS	external bridges untouched

8 What this does not prove

It does not prove or disprove a Carleson estimate for infinitely many selected windows. It does not prove a source-to-border large sieve. It does not turn regression into a bound. It does not infer stability from a strict SCRAMBLE band that excludes supercritical eigenvalues. It does not alter any Lean theorem or any claim status. No RH claim.